2016 IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC)
June 1-4, 2016, Busan, Korea

Torque Ripple Reduction of BLDC motor using Predicted Current Control

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***Abstract*— This paper presents a simple predicted current control scheme for BLDC(Brushless DC) motor to reduce the torque ripple in the commutation and conduction region.**

In the proposed control scheme, the inverse function of back emf is used to consider the back emf characteristic in the conduction region. In the commutation region, a predicted current of incoming and outgoing phase are used to calculate the slope and variation of commutated current. The proper duty ratio to keep the constant current is supplied during commutate to reduce the torque ripple. This method is realized by the precise prediction of the phase current, continuous back emf by the look-up table according to the rotor position and measured speed.

Finally, experiment is performed and the experiential results show that the proposed control scheme can well keep the constant torque during the conduction and commutation region.

Keywords— BLDC(Brushless Direct Current) motors, Torque ripple reduction, Predicted Current Control

I. INTRODUCTION

BLDC(Brushless DC) motor are widely used in various applications such as robot, electric bicycle, wheelchairs and home appliances due to the high power density and efficiency with an excellent maintenance characteristic[1-6]. Furthermore, small BLDC motor are much interested in the un-manned aero vehicles such as a drone. The usage of high power with high speed BLDC motor s are dramatically increased in these markets.

Although BLDC motors have an excellent torque characteristics, the output torque has torque ripple due to the non-linear back-EMF of the motor and the commutation current ripple. As well known, the back-EMF of the motor is affected by the characteristics of the permanent magnet and stator, and has high order harmonics due to the slot effect and magnetization direction of the magnet with concentrated phase windings. These harmonics make torque ripple with a constant phase current in the conduction region. Furthermore the current ripple in the commutation region where the conducting phase changing region, makes additional torque ripple. [1-3].

In this paper, a separated current control scheme as the conduction mode and the commutation mode to reduce the torque ripple of BLDC motor is presented. Because, the main torque ripple sources are different in each control period, the proposed control modes can effectively reject the proper torque ripple sources. The proposed conduction mode controller uses an inversed back-EMF function to consider the non-linear back-EMF effect of the BLDC motor. To have same commutation current variation, a proper turn-on time of the commutate phases is calculated based on the voltage equation of BLDC motor .

In the calculation, switching voltage by the turn-on time and the estimated back emf are used to predict the next sample current.

The proposed current control scheme is verified by the practice experiment.

II. CONDUCTION TORQUE RIPPLE OF BLDC MOTOR

The output torque of BLDC motor is defined by the back EMF of the motor and phase current. Fig. 1 shows the ideal back EMF of BLDC motor and phase current. In the conduction region which is separated as $\frac{\pi}{3}$ region, the output torque of the motor can be produced as follow. If the back emf has non-ideal waveform, proper phase current according to the back emf should be controlled to keep a constant output torque[4].

$$T_m = \frac{e_{as}\cdot i_{as}+e_{bs}\cdot i_{bs}+e_{cs}\cdot i_{cs}}{\omega_{rm}} \quad (1)$$

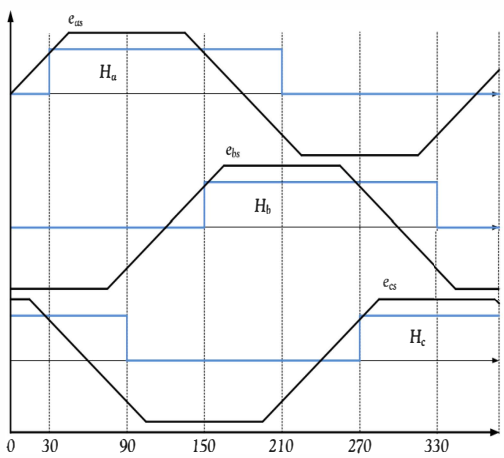

Fig. 1. Ideal Back emf and phase current of BLDC motor

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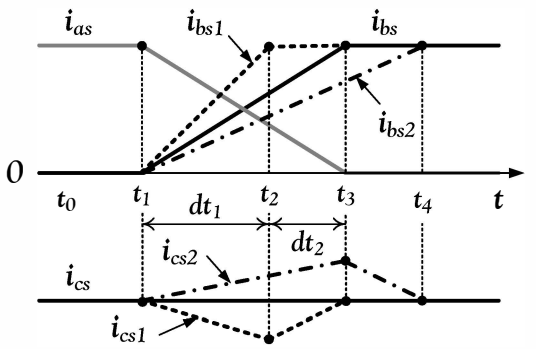
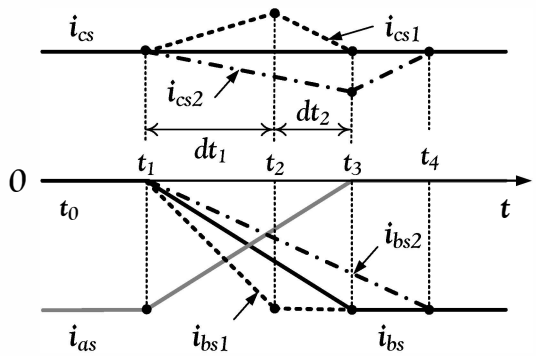
III. COMMUTATION TORQUE RIPPLE OF BLDC MOTOR

Fig. 2 shows phase current waveforms of a BLDC motor during the commutation region in the positive commutation and the negative commutation respectively. As previously stated, the outgoing phase current cannot be dropped to zero at t_1 point shown in Fig. 2. Similarly, the incoming phase current cannot be built up to the demanded reference DC value at the same time due to the winding inductance and supplied voltage of the motor. When the incoming phase current is built up faster than the outgoing phase current at t_2 point shown in Fig. 2, the phase current $i_{cs}1$ of the remaining phase has spike shape current ripple [2]. On the contrary, the incoming phase current is built up slower than the outgoing phase current at t3 point, the phase current $i_{cs}2$ of the remaining phase has dip shape current ripple [5-6].

In order to keep the constant current for the constant output torque during commutation region, the current slope of the incoming phase and outgoing phase should be same.



(a) Commutated current is positive



(b) Commutated current is negative

Fig. 2. Current behaviors during commutation of BLDC motor

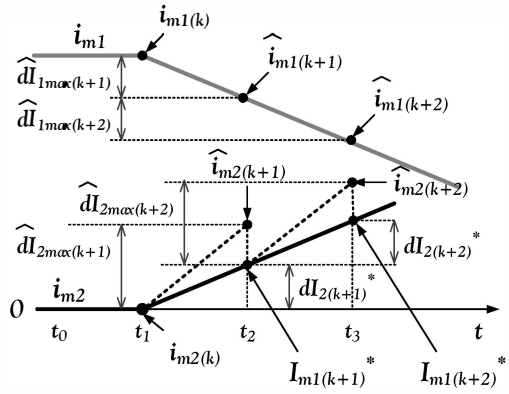
IV. THE PROPOSED CURRENT CONTROL SCHEME

In order to make constant torque during commutation region, the optimized turn-on time is calculated in this paper. Based on the instantaneous voltage equation of a BLDC motor , the incoming and outgoing phase's voltage equations can be written as follows.

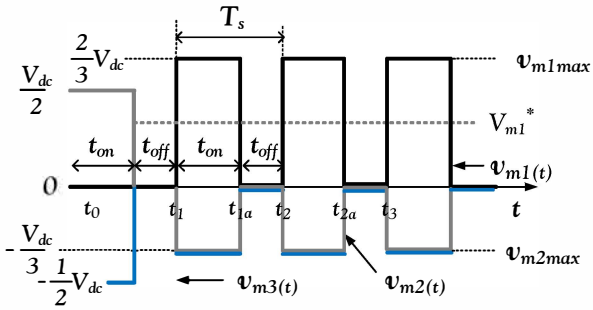
$$v_{m1} = R_{m1} \cdot i_{m1} + L_{m1} \cdot \frac{di_{m1}}{dt} + e_{m1} \tag{2}$$

$$v_{m2} = R_{m2} \cdot i_{m2} + L_{m2} \cdot \frac{di_{m2}}{dt} + e_{m2} \tag{3}$$

In (2) and (3), R_{m1} and R_{m2} are phase winding resistances of outgoing and incoming phase. e_{m1} and e_{m2} are instantaneous back emf of each phases.



(a) Commutation current behaviors



(b) Switching voltages during commutation region

Fig. 3. Current behaviors and switching voltages in commutation region

Fig. 3 shows the current behavior during commutation region with switching voltages according to the PWM duty ratio. In Fig. 3, the switching voltages are determined by the current states. In the conduction region, two phases are activated. So, supplied winding voltages are determined as follows.

$$\begin{aligned} v_{m1(k)} &= \frac{1}{2} V_{dc} \cdot t_{on(k)} \\ v_{m2(k)} &= \frac{2}{3} V_{dc} \cdot t_{on(k)} \\ v_{m3(k)} &= 0 \end{aligned} \tag{4}$$

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However, switching voltages are changed in the commutation region according to the unchanged phase current state. If the unchanged phase current is negative, switching voltages are defined as follows.

$$v_{m1(k)} = -\frac{1}{3}V_{dc} \cdot t_{on(k)}$$
$$v_{m2(k)} = \frac{2}{3}V_{dc} \cdot t_{on(k)} \qquad (5)$$
$$v_{m3(k)} = -\frac{1}{3}V_{dc} \cdot t_{on(k)}$$

Where, $t_{on(k)}$ is defined as turn-on time in the current sampling period. In the proposed current control scheme, same duty ratio is supplied in two phase windings. Current variations by switching voltages can be derived as follows.

$$di_{m1(k+1)} = \frac{1}{L_m} \cdot \left(v_{m1(k)} - R_m \cdot i_{m1(k)} - e_{m1(k+1)}\right) \cdot T_s \qquad (6)$$
$$di_{m2(k+1)} = \frac{1}{L_m} \cdot \left(v_{m2(k)} - R_m \cdot i_{m2(k)} - e_{m2(k+1)}\right) \cdot T_s \qquad (7)$$

In order to calculate the current variation in every sampling period, the continuous back emf should be known. The back emf of the BLDC motor can be modelized according to the motor speed and rotor position. In this paper, the rotor position of the motor is estimated by the hall sensor signal and calculated motor speed by the hall sensor signal intervals.

$$\hat{\theta}_{re(k)} = \theta_{re(n)} + \omega_{re(n)} \cdot (k-n) \cdot T_s \qquad (8)$$

In equation (8), $\omega_{re(n)}$ is the calculated speed at n-step when the hall sensor signal is changed point. And $\theta_{re(n)}$ is the prefixed rotor position which is matched to the hall sensor signal changing point.

From the estimated rotor position, the back emf of motor is estimated as follow.

$$\hat{e}_{m1(k+1)} = f_e\left(\omega_{re(k)}, \hat{\theta}_{re(k+1)}\right) \qquad (9)$$

In this paper, the back emf of the motor is pre-measured according to the motor speed and rotor position, then the position data is normalized by the motor speed and rotor position. Fig. 4 shows the estimation block of a continuous back emf for the proposed current control scheme.

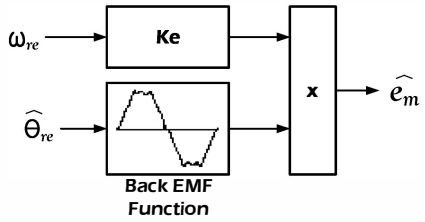

Fig. 4. The estimation block of a continuous back emf

From equation (8) and (9), we can derive the current variation of the outgoing and incoming phase.

$$\widehat{di}_{m1(k+1)} = \frac{1}{L_m} \cdot \left(v_{m1(k)} - R_m \cdot i_{m1(k)} - \hat{e}_{m1(k+1)}\right) \cdot T_s \qquad (10)$$
$$\widehat{di}_{m2(k+1)} = \frac{1}{L_m} \cdot \left(v_{m2(k)} - R_m \cdot i_{m2(k)} - \hat{e}_{m2(k+1)}\right) \cdot T_s \qquad (11)$$

In order to reduce the torque ripple, current variation $\widehat{di}_{m1(k+1)}$ is same to $\widehat{di}_{m2(k+1)}$. In this condition, the proper turn-on time of the next sampling period in the commutation region can be determined as follow with equation (5) .

$$t_{on}^{*} = \frac{3 \cdot \{R_m \cdot (i_{m2(k)} + i_{m1(k)}) + (\hat{e}_{m2(k+1)} + \hat{e}_{m1(k+1)})\}}{V_{dc}} \qquad (12)$$

When the un-changed phase current is positive, the proper turn-on time is determined as follow.

$$t_{on}^{*} = \frac{-3 \cdot \{R_m \cdot (i_{m1(k)} + i_{m2(k)}) + (\hat{e}_{m1(k+1)} + \hat{e}_{m2(k+1)})\}}{V_{dc}} \qquad (13)$$

Fig. 5 shows the block diagram of the proposed current control scheme. In the proposed control scheme, the phase current is controlled by the speed or torque controller.

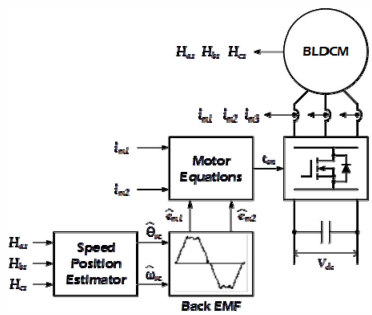

Fig. 5. Block diagram of the proposed control scheme

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V. EXPERIMENTAL RESULTS

In order to verify the proposed current control scheme practice experiment are implemented with 1.2kW BLDC motor. Table I shows the motor specifications.

TABLE I. MOTOR SPECIFICATIONS

Maximum Power	1.2kW
Rated Voltage	48V
Maximum current	25A
Maximum Torque	4.58Nm
Speed	2500rpm

Fig. 6 and Fig. 7 show experimental results of the conventional current control scheme and the proposed current control scheme during conduction region at 500rpm. In experiment results, the conduction torque ripples are almost same. But the commutation torque ripple of the conventional control scheme is higher than the proposed method.

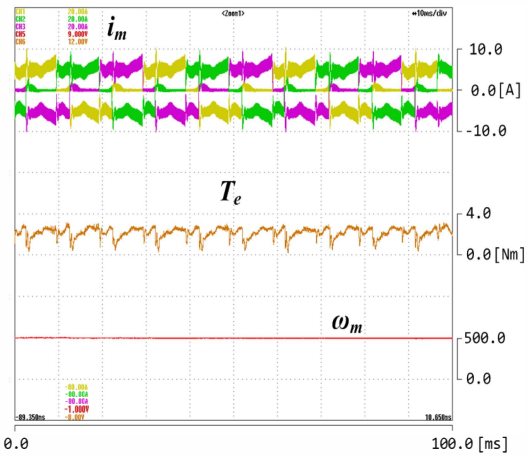

Fig. 6. Experiment result of the conventional current control during conduction region at 500rpm

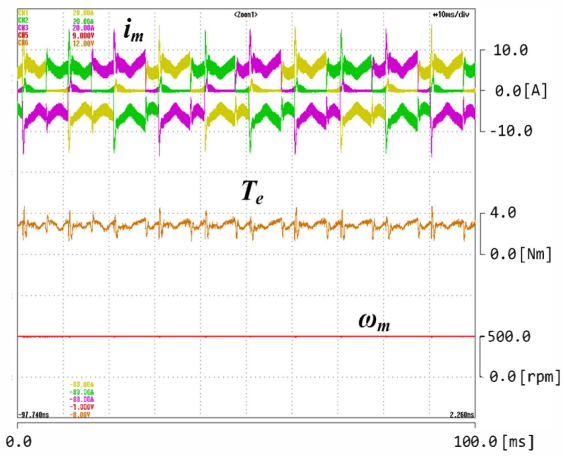

Fig. 7. Experiment results of the proposed method during conduction region at 500rpm

Fig. 8 and Fig. 9 show experimental results of the conventional current control scheme and the proposed current control scheme during commutation region. As a result, the non-commutated phase current has ripple during commutation. This current ripple makes an additional torque ripple during commutation region And the supplied voltages are not controlled to keep the same current slope.

In the proposed current control scheme shown in Fig. 9, the rising current of the incoming phase and the distinguished current of the outgoing phase are controlled by the PWM duty ratio keep the same current variation in every sampling period.

So the commutation times of two phases are almost same. As a result, the output torque is constant during commutation region except the torque ripple by the PWM switching.

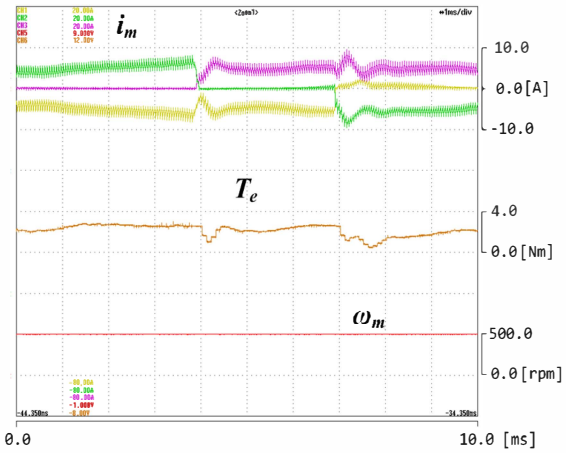

Fig. 8. Experiment result of the conventional current control during commutation region at 500rpm

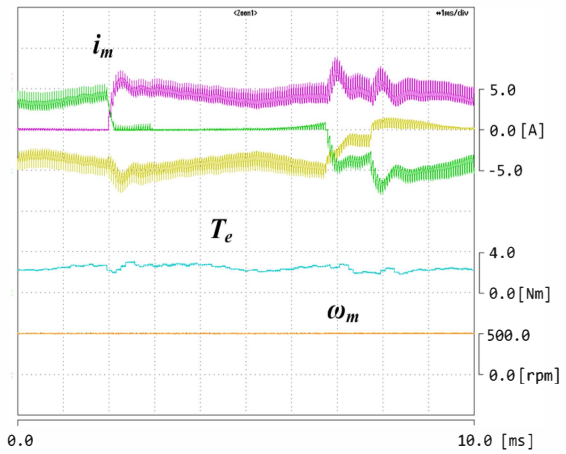

Fig. 9. Experiment results of the proposed method during commutation region at 500rpm

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Fig. 10 and Fig. 11 show the detailed comparison results of the proposed current control scheme for during the conduction and commutation reign at 500rpm.

As shown in Fig. 10, Phase current consider inverse function of back emf. So reduce the torque ripple. And Fig. 11 show the rising current of the incoming phase winding and the distinguished current of the outgoing phase are controlled by the PWM duty ration to keep the same current variation in every sampling period. So the commutation times of two phases are almost same. As a experiment result, the output torque is constant during conduction and commutation region reduce the torque ripple.

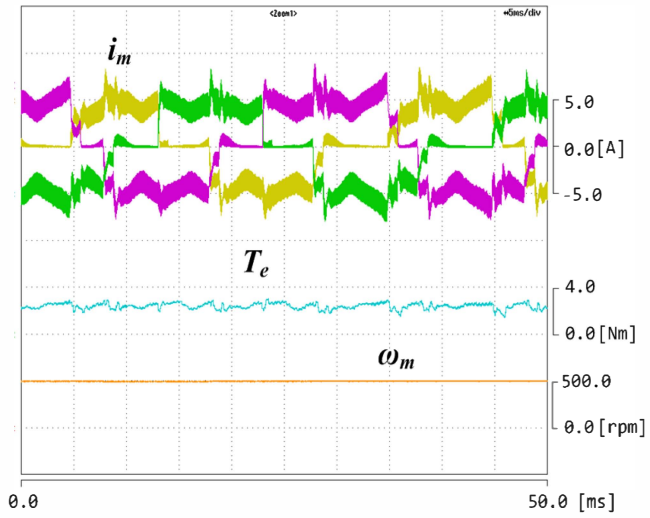

Fig. 10. Experiment results of the proposed method during conduction region at 500rpm

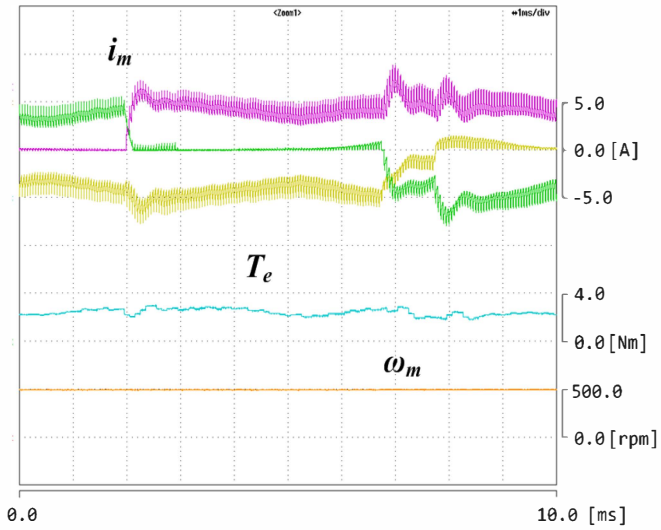

Fig. 11. Experiment results of the proposed method during commutation region at 500rpm

VI. Conclusions

This paper presents a predicted current control of BLDC motor to reduce torque ripple. The proposed method consider a non-ideal back emf and calculate turn-on time of PWM method to make same current inclining slope of the incoming phase and declining slope of the outgoing phase current. From this simple PWM switching, the torque ripple can be reduced.

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